

Education

University of Connecticut

Sep 2024 – May 2028 (Expected)

*PhD Candidate in Statistics, Department of Statistics**Storrs, CT, U.S.*

- Courses: Design of Experiments, Statistical Inference, Linear Models

Renmin University of China

Sep 2022 – Jun 2024

*Master of Science in Applied Statistics, Institute of Statistics and Big Data**Beijing, China*

- Courses: Clinical Trial Statistical Methodology, Causal Inference, Survival Analysis

Xi'an Jiaotong University

Sep 2018 – Jun 2022

*Bachelor of Economics in Finance, School of Economics and Finance**Xi'an, China*

- Courses: Probability Theory, Mathematical Statistics, Econometrics

Work Experience

Astellas Pharma

May 2026 – Aug 2026

*Biostatistician Intern**Remote*

Closed-Form Evaluation of Correlated PFS and OS Group Sequential Designs

- Developed an event-driven group sequential design for correlated progression-free survival (PFS) and overall survival (OS) under the Fleischer model, supporting efficacy monitoring, futility stopping, and piecewise uniform accrual.
- Derived a closed-form joint correlation matrix for sequential PFS and OS log-rank statistics, reported stopping boundaries on z-score, p-value, and hazard-ratio scales, calculated marginal power and gatekeeping joint power and validated the results against Monte Carlo simulations ($< 10^{-4}$ absolute discrepancy).
- Implemented the methodology in an R package and Shiny application, deployed via Posit Connect and used by Astellas statisticians for trial-design comparisons; presented the work at an internal statistics meeting and incorporated user feedback to improve the tools' functionality and usability.

Research Experience

Department of Statistics, UConn

Sep 2025 – Present

*Dissertation Research, Advised by Prof. Haiying Wang and Prof. Jun Yan**Storrs, CT, U.S.*

Rerandomization Methods for Experimental Design

- Conducting a comprehensive review of rerandomization methodologies, synthesizing core theoretical developments, and identifying open research questions in the field.
- Investigating the adverse impact of high-dimensional covariates on rerandomization, including strategies to mitigate high-dimensional imbalance and analysis of the associated statistical properties.
- Developing principled variable selection strategies under rerandomization to identify influential covariates while maintaining design validity and statistical efficiency.
- Building a high-performance R package for rerandomization-based experimental design and analysis, incorporating recent theoretical advances with scalable and user-friendly implementations.

Department of Public Health Sciences, UConn Health

Sep 2025 – Dec 2025

*Research Assistant**Farmington, CT, U.S.*

Longitudinal Analysis of Depression Outcomes in Large-Scale Medicare Claims Data

- Processed and linked over 3 billion Medicare claims records using `fst` and `data.table` to construct a longitudinal cohort of more than 2 million beneficiaries with depression, integrating claims history, chronic-condition indicators, and demographic covariates.
- Used Cox proportional hazards models with annual follow-up intervals to estimate associations of clinical and demographic factors with time to depression remission, defined as the first annual assessment at which a beneficiary no longer met the study's depression criterion based on the depression score.

Center for Population Health, UConn Health

Jan 2025 – Jun 2025

*Research Assistant**Farmington, CT, U.S.*

Survival Analysis of Suicide Risk Using Clinical Data

- Analyzed over 5 million longitudinal clinical encounter records and engineered predictors from ICD-9/10 diagnosis histories, Charlson comorbidity scores, medication exposures, and demographic characteristics.

- Used Weibull and Cox proportional hazards models to identify clinical and demographic factors associated with time to the first suicide-related event; used logistic regression of the binary suicide indicator as a complementary analysis to assess the consistency of identified risk factors.

Connecticut Children's Medical Center

Research Assistant

Aug 2024 – Jun 2025

Hartford, CT, U.S.

Time-Series Forecasting of Emergency Department Patient Volume

- Preprocessed and visualized emergency department census data from Connecticut Children's Medical Center, using heatmap calendar plots to reveal key periodic patterns and temporal trends in patient flow.
- Collected external covariates (temperature, air pollution, water quality) from public APIs, integrated them with census data, and performed data merging for downstream modeling and analysis.
- Utilized Fourier series to construct a harmonic dynamic regression model to estimate different periodic cycles, combined with an ARIMAX framework to model non-periodic short-term variations, achieving effective forecasting performance.

Institute of Geriatric Medicine, Beijing Hospital

Research Assistant

Oct 2022 – May 2023

Beijing, China

Change-Point Analysis of Physiological Changes in Aging Populations

- Streamlined and preprocessed population data from Beijing Hospital, and utilized line plots and cubic spline regression plots to identify significant change points in various physiological function variables.
- Estimated knots in linear spline models using the Newton-Raphson algorithm, specifically tailored to capture critical shifts related to the aging process.
- Developed advanced plotting functions to visually validate results and engaged in thorough consultations with hospital doctors, ensuring a robust integration of statistical findings with clinical experiences.

Institute of Statistics and Big Data, Renmin University of China

Dissertation Research, Advised by Prof. Yang Liu

Mar 2023 – Jun 2024

Beijing, China

Covariate-Adaptive Randomization for Multi-Arm Experiment with Unequal Allocations

- Developed improved covariate-adaptive randomization procedures for multi-arm clinical trials, proposing novel randomization schemes that enhance covariate balance while preserving randomization integrity.
- Studied theoretical properties of CAR procedures by deriving asymptotic distributions of covariate imbalances under general multi-arm designs with unequal allocation ratios.

Teaching Experience

Institute of Statistics and Big Data, Renmin University of China

Teaching Assistant for Course Clinical Trial Methodology

Sep 2023 – Jan 2024

Beijing, China

Publications

- Xingjian Ma, Robert H. Aseltine, Tze Chiam, Richelle deMay, Yingfa Xie, and Jun Yan. Forecasting Emergency Department Census: Leveraging Periodic Patterns and External Covariates. (*Major Revision*)
- Xie, Y., Aseltine, R. H., Chiam, T., deMayo, R., Huang, Y., Ma, X., & Yan, J. (2025). Modeling Emergency Department Volume-at-Risk Using a Discrete Extreme Value Approach. *The Journal of the Royal Statistical Society: Series C*. (*Major Revision*)
- Xingjian Ma and Yang Liu. Balancing Observed and Unobserved Covariates with Multi-arm Covariate-Adaptive Randomization for Unequal Allocations. (*In preparation*)

Projects

CorrSurvGSD: R Package for Correlated PFS and OS Group Sequential Designs | R, Shiny 2026

- Built an R package with Shiny app that enables statisticians to specify two-arm sample sizes and median survival times, accrual rates, and group sequential monitoring settings, including efficacy and futility rules, for correlated PFS/OS designs; generates calendar cutoffs, stopping boundaries, marginal power, and gatekeeping joint power.

Parkinson's Disease Progression Prediction | Python (PyTorch)

May 2023 – Jun 2023

- Engineered time-aware clinical features from high-dimensional proteomic data, reducing 1,200+ variables to 25 informative predictors, enhancing model interpretability and performance.
- Developed a hybrid model combining LightGBM (multi-class classification) and a neural network and applied strict time-series cross-validation to ensure robust generalization on future patient visits.

Covariate-Adjusted Randomization in Group Sequential Design | R, Rcpp

Feb 2023 – Dec 2023

- Incorporated covariates in a multivariate regression analysis to aid in estimating the variance of treatment effects, further optimizing the efficiency of interim analysis and reducing the expected sample size.
- rerand: R package for Rerandomization** | *R, Rcpp* **Oct 2025 – Dec 2025**
- Developed an R package for rerandomization experiments, allowing users to select covariates separately for treatment assignment and analysis; provides difference-in-means, ANCOVA, and Lin estimators with conservative variance estimates, and Gaussian-mixture-based confidence intervals and hypothesis tests.
 - Implemented efficient Rcpp routines for large-scale rerandomization, achieving a 5x speedup in generating treatment assignments over pure R.
- Deep Reinforcement Learning** | *Python (PyTorch)* **Oct 2022 – Dec 2022**
- Developed AI-driven algorithms employing CNN architectures (e.g., AlexNet, VGGNet, ResNet) for enhanced interactive game image analysis, incorporating data augmentation and transfer learning techniques to boost model robustness.
 - Utilized reinforcement learning strategies, specifically the Q-learning algorithm, to continuously train and refine the model based on dynamic game scenarios, resulting in improved decision-making and adaptability.

Honors & Awards

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| • Outstanding Graduate of Renmin University of China | Apr 2024 |
| • Graduate Excellent First Class Scholarship, Renmin University of China | Oct 2023 |
| • Honorary Title of Outstanding Student Cadre, Renmin University of China | Oct 2023 |

Technical Skills

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- Programming:** R (dplyr, tidyverse, ggplot2); Python (Numpy, Pandas, PyTorch); Rcpp; SAS; Git; Linux
- Languages:** Chinese(Native); English(Fluent - TOEFL: 100 (R28 L26 S22 W24))